

Automated Engineering Report: Neural Network Engine

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Generated on: \$(date)

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1 1. Introduction

This documentation is automatically generated by analyzing the source code. It reflects the current state of the architecture.

2 2. Automated Class Analysis

2.1 Class: Layer

Found in `Layer.h`. This class is a core component of the engine.

2.1.1 Public Interface:

- `Matrix forward(const Matrix& input)`
- `Matrix backward(const Matrix& outputGradient, double learningRate)`

2.2 Class: Matrix

Found in `Matrix.h`. This class is a core component of the engine.

2.2.1 Public Interface:

- `double& operator()(int row, int col)`
- `void randomize(double min, double max)`
- `void fillZero()`
- `void applyReLU()`

2.3 Class: Network

Found in `Network.h`. This class is a core component of the engine.

2.3.1 Public Interface:

- `Matrix feedForward(const Matrix& input)`
- `void backpropagate(const Matrix& input, const Matrix& target)`
- `void train(Matrix* inputs, Matrix* targets, int numSamples, int epochs)`

3 3. Mathematical Operations

The engine currently supports: - **Forward Propagation:** $Z = W \cdot X + B$ - **ReLU Activation:** $f(z) = \max(0, z)$ - **Backpropagation:** Gradient descent optimization.